

Lab Experiment 10: Working with Files

Objective

To perform basic file-handling operations in Linux, including:

1. **Creating and editing text files** using commands and editors.
2. **Archiving, compressing, unpacking, and uncompressing files** using `tar`, `star`, `gzip`, and `bzip2`.
3. **Creating, deleting, copying, and moving files and directories.**

Background

In Linux, everything is considered a **file** — whether it's a text document, image, program, or directory. Understanding how to create, modify, and manage files is a **core skill** for every Linux user.

II. Create and Edit Text Files

Before learning compression and file management, it's essential to know **how to create and modify text files** in Linux.

Common Tools for File Creation and Editing

Tool	Purpose	Suitable For
<code>touch</code>	Creates an empty file instantly.	Beginners, quick file creation
<code>cat</code>	Creates or views small text files.	Short files and notes
<code>nano</code>	Simple text editor with menus.	Beginners
<code>vi/vim</code>	Advanced editor for experienced users.	Professionals, coders

Step-by-Step Commands

1 Create Empty Files using `touch`

```
touch sample.txt
touch report.txt
```

Check created files:

```
ls
```

2 Create and Write Content using cat

```
cat > notes.txt
```

Type content, then press **Ctrl + D** to save.

View the file:

```
cat notes.txt
```

3 Append Text to an Existing File

```
cat >> notes.txt
```

4 Create or Edit File using nano

```
nano notes.txt
```

Save: **Ctrl + O**, Exit: **Ctrl + X**

Outcome of Section II

Students will be able to:

- Create files using **touch**.
- Create, view, and edit text files using **cat**, **nano**.
- Append additional text to existing files.

I. Archive, Compress, Unpack, and Uncompress Files

Terminologies

Term	Description	When to Use
Archiving	Combines multiple files/folders into one single file.	Backups or transfers.
Compression	Reduces file size using algorithms.	Save disk space.
Unpacking	Extracts files from an archive.	Access archived files.
Uncompressing	Restores compressed files.	Use compressed data.
tar	Tool for archiving multiple files.	Common archiver.
star	Standard TAR with extra features.	Advanced archiving.
gzip	Fast compression utility.	Single files quickly.
bzip2	Stronger, slower compression utility.	Maximum space saving.

Step-by-Step Commands

1 Prepare Sample Files

```
mkdir ArchiveDemo
cd ArchiveDemo
touch file1.txt
touch file2.txt
```

2 Archive Files using tar

```
tar -cvf myfiles.tar file1.txt file2.txt  (#Archives)
tar -xvf myfiles.tar  (#Extract)
```

3 Archive + Compress (gzip / bzip2)

gzip:

```
tar -czvf myfiles.tar.gz file1.txt file2.txt  (#Archives)
tar -xzvf myfiles.tar.gz  (#Extract)
```

bzip2:

```
tar -cjvf myfiles.tar.bz2 file1.txt file2.txt  (#Archives)
tar -xjvf myfiles.tar.bz2  (#Extract)
```

4 Compress / Uncompress Single Files

gzip:

```
gzip file1.txt  (#Compress)
gunzip file1.txt.gz  (#Uncompress)
```

bzip2:

```
bzip2 file2.txt  (#Compress)
bunzip2 file2.txt.bz2  (#Uncompress)
```

5 Using star

```
star -c -f=backup.star file1.txt file2.txt
star -x -f=backup.star
```

Outcome of Section I

Students will be able to:

- Use `tar`, `star`, `gzip`, `bzip2`.
- Create archives and compressed files.
- Extract or uncompress archives.

III. Create, Delete, Copy, and Move Files and Directories

Basic File and Directory Commands

Operation	Command	Example
Create an empty file	<code>touch filename</code>	<code>touch demo.txt</code>
Create a directory	<code>mkdir dirname</code>	<code>mkdir work</code>
Copy file	<code>cp source destination</code>	<code>cp demo.txt work/</code>
Copy directory	<code>cp -r dirname destination</code>	<code>cp -r work backup/</code>
Move or rename file	<code>mv oldname newname</code>	<code>mv demo.txt info.txt</code>
Delete file	<code>rm filename</code>	<code>rm info.txt</code>
Delete empty directory	<code>rmdir dirname</code>	<code>rmdir work</code>
Delete non-empty directory	<code>rm -r dirname</code>	<code>rm -r backup</code>

Practice Tasks

1. Create a folder `TestData` and two files inside it using `touch`.
2. Write content into the files using `cat` or `nano`.
3. Copy both files into a directory named `Backup`.
4. Rename one file.
5. Compress the files using `gzip`.
6. Delete the original folder using `rm -r`.
7. Extract the compressed files and verify their content.

Summary

Section	Topic	Skills Learned
II	Create and Edit Text Files	Create and modify text files using touch, cat, nano, and vi.
I	Archive and Compress Files	Use tar, star, gzip, bzip2 for backup and compression.
III	File and Directory Management	Create, copy, move, and delete files/directories.